Junshen Xu

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Education

- **Massachusetts Institute of Technology** — Cambridge, MA
  PhD, Electrical Engineering and Computer Science — *Sep 2018 – May 2023*
- **Massachusetts Institute of Technology** — Cambridge, MA
  MS, Electrical Engineering and Computer Science — *Sep 2018 – May 2020*
- **Tsinghua University** — Beijing, China
  BSc, Engineering Physics — *Aug 2014 – Jul 2018*

Experience

- **AWS** — Boston, MA
  Senior Applied Scientist, Security Analytics and AI Research — *Jun 2023 - present*
  - **Autonomous Co-evolving Security Agents**: Built autonomous multi-agent systems for security investigations in AWS cloud environments, leveraging adversarial co-evolution to continuously improve reasoning, tool use, and investigation quality.
  - **Foundation Models for Threat Intelligence**: Developed domain-specific LLMs for cybersecurity threat intelligence, incorporating RAG and reinforcement learning to improve reasoning over evolving threat knowledge.
  - **Multi-domain Threat Detection**: Designed and deployed a large-scale data processing and ML pipeline that correlates signals across heterogeneous security domains, reducing false positives over 98%.
- **Meta** — New York, NY
  Machine Learning Engineer Internship, Instagram Reels Core Ranking — *May 2022 - Aug 2022*
  - **Mitigate Short Video Bias in Reels**: Designed and developed new signals to mitigate short video bias in Reels ecosystem. Trained and deployed deep learning ranking models with new signals. Performed A/B tests to evaluate the new models. The proposed signals mitigated the short video bias (+20% long video play), increased user time spent (+0.5%) and achieved gains in user sessions (+0.1%))
- **Google** — New York, NY
  Software Engineer Internship, Adwords Insights Team — *Jun 2021 - Aug 2021*
  - **Negative Trends Mitigation**: Mitigated negative reports of insight trends with a new keyword classification method in search ads.
  - **Cold Start Detection**: Proposed and Implemented a novel algorithm to detect the cold start periods of machine learning based bidding strategies in Adwords.
- **United Imaging Intelligence America** — Cambridge, MA
  Research Scientist Internship — *Jun 2020 - Aug 2020*
  - **Multi-modal Registration**: Developed a deep learning based multi-modal registration framework with multi-modal GAN and neural ODE.
  - **Digitally Reconstructed Radiograph**: Implemented a differentiable DRR rendering operator on GPU with CUDA C++. Developed a 2D/3D registration method for DRR and CT with deep learning.
- **Massachusetts Institute of Technology** — Cambridge, MA
  Research Assistant — *Sep 2018 - May 2023*
  - **2D-to-3D Reconstruction**: Proposed a robust and efficient method for reconstructing/rendering a 3D volume from multiple motion-corrected 2D slices using Transformers and Implicit Neural Representations.
  - **Image Quality Assessment (IQA)**: Developed a semi-supervised learning method for IQA and built a real-time IQA system on a MR scanner.
  - **3D Pose Estimation Pipeline**: Developed a automatic pipeline for 3D fetal MRI data processing, including image super-resolution, keypoint detection (pose estimation), and motion analysis.

Selected Publications

[1] **Xu, J.**, Zhang, J., Fan Y., Multi-Domain Marker Aggregation for Threat Detection in Cloud Environments. WWW 2026.

[2] Zhang, J., **Xu, J.**, Can, B., Fan Y., REACT: Residual-Adaptive Contextual Tuning for Fast Model Adaptation in Threat Detection. WWW 2025.

[3] Zhang, M., **Xu, J.**, Arefeen, Y., Adalsteinsson, E. Zero-Shot Self-Supervised Joint Temporal Image and Sensitivity Map Reconstruction via Linear Latent Space. In Medical Imaging with Deep Learning. MIDL 2024.

[4] **Xu, J.**, Moyer, D., Gagoski, B., Iglesias, J. E., Grant, P. E., Golland, P., Adalsteinsson, E. NeSVoR: Implicit Neural Representation for Slice-to-Volume Reconstruction in MRI. IEEE TMI (2023).

[5] Vasung, L., **Xu, J.**, Abaci-Turk, E., Zhou, C., Holland, E., Barth, W. H., et al. Cross-sectional Observational Study of Typical in-utero Fetal Movements using Machine Learning. Developmental Neuroscience (2023).

[6] Arefeen, Y., **Xu, J.**, Zhang, M., Z Dong, F Wang, J White, et al. Latent Signal Models: Learning Compact Representations of Signal Evolution for Improved Time-Resolved, Multi-Constrast MRI. MRM (2023).

[7] Liu, C., Karani, N., Abulnaga, M., **Xu, J.**. Consistency Regularization Improves Placenta Segmentation in Fetal EPI MRI Time Series. MICCAI PIPPI 2023.

[8] **Xu, J.**, Moyer, D., Grant, P. E., Golland., P., Iglesia, J. E., Adalsteninsson, E. SVoRT: Iterative Transformer for Slice-to-Volume Registration in Fetal Brain MRI. MICCAI 2022 (oral presentation).

[9] Gagoski, G., **Xu, J.**, Wighton, P., Tisdall, D., Frost, R., Lo, W., et al. Automated detection and reacquisition of motiondegraded images in fetal HASTE imaging at 3T. MRM (2022)

[10] **Xu, J.**, Turk, E. A., Grant, P. E., Golland., P., Adalsteinsson, E. STRESS: Super-Resolution for Dynamic Fetal MRI using Self-Supervised Learning. MICCAI 2021.

[11] **Xu, J.**, Adalsteinsson, E. Deformed2Self: Self-Supervised Denoising for Dynamic Medical Imaging. MICCAI 2021.

[12] **Xu, J.**, Chen, E.Z., Chen, X., Chen, T. and Sun, S. Multi-scale Neural ODEs for 3D Medical Image Registration. MICCAI 2021.

[13] **Xu, J.**, Lala, S., Gagoski, B., Turk, E. A., Grant, P. E., Golland., P., Adalsteinsson, E. Semi-Supervised Learning for Fetal Brain MRI Quality Assessment with ROI consistency. MICCAI 2020.

[14] Zhang, M., **Xu, J.**, Turk, E. A., Grant, P. E., Golland., P., Adalsteinsson, E. Enhanced Detection of Fetal Pose in 3D MRI by Deep Reinforcement Learning with Physical Structure Priors on Anatomy. MICCAI 2020.

[15] **Xu, J.**, Zhang, M., Turk, E. A., Grant, P. E., Golland., P., Adalsteinsson, E. 3D Fetal Pose Estimation with AdaptiveVariance and Conditional GenerativeAdversarial Network. MICCAI PIPPI 2020.

[16] **Xu, J.**, Gong, E., Ouyang, J., Pauly, J., Zaharchuk, G.Ultra-low-dose 18F-FDG brain PET/MR denoising using deep learning and multi-contrast information. SPIE Medical Imaging (2020) (oral presentation)

[17] **Xu, J.**, Zhang, M., Turk, E. A., Zhang, L., Grant, P. E., Ying, K., et al. Fetal Pose Estimation in Volumetric MRI Using a 3D Convolution Neural Network. MICCAI 2019.

[18] Jiang, M., Zhang, Y., **Xu, J.**, Ji, M., Guo, Y., Guo, Y., et al. Assessing EGFR gene mutation status in non-small cell lung cancer with imaging features from PET/CT. Nuclear medicine communications (2019)

[19] Chen, K., Gong, E., Macruz, F., **Xu, J.**, Boumis, A., Khalighi, M., et al. Ultra-low-dose 18F-florbetaben Amyloid PET Imaging using Deep Learning with Multi-contrast MRI Inputs. Radiology (2018).

[20] **Xu, J.**, Gong, E., Pauly, J., & Zaharchuk, G. 200x Low-dose PET reconstruction using deep learning. arXiv preprint arXiv:1712.04119.